

Gram-Negative Lipoprotein-Peptidoglycan Tether Remodeling in *Pseudomonas putida* KT2440

Commissioned Module/Pathway/Taxon Curation Review

Taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI:txid160488; proteome UP000000556) **Local bucket reviewed:** KEGG [ppu00550](#) "Peptidoglycan biosynthesis" (10 primary genes; module area *lipid_cell_envelope_metabolism*) **Module under review:** reversible covalent attachment of an outer-membrane lipoprotein to peptidoglycan — tether **formation** (ErfK-family L,D-transpeptidase) + tether **hydrolysis** (YafK/LdtF/DpaA-family hydrolase)

Summary

The single most important curation conclusion of this review is a **scope mismatch between the module and its assigned candidate list**. The commissioned module describes reversible covalent attachment of an outer-membrane (OM) lipoprotein to peptidoglycan (PG): an ErfK/YkuD-family **L,D-transpeptidase** anchors a Braun-lipoprotein-like substrate, and a YafK/LdtF/DpaA-family **cysteine hydrolase** can release it. **None** of the 23 candidate genes drawn from KEGG [ppu00550](#) encode either activity. All 23 are canonical PG-biosynthesis and cell-division proteins (Mur ligases, penicillin-binding proteins, SEDS glycosyltransferases, D,D-carboxypeptidases, undecaprenyl-phosphate enzymes) that perform glycan polymerization and 4→3 D,D-crosslinking — reactions the module scope explicitly **excludes**. The [ppu00550](#)→module link is therefore a **boundary error**; the module should be detached and marked [module_needs_revision](#).

The module's actual enzymes lie outside the candidate list. KT2440 encodes exactly two proteins carrying the diagnostic YkuD/ErfK L,D-transpeptidase domain (Pfam **PF03734** / InterPro IPR005490): **PP_2320** (Q88KH0) and **PP_1451** (Q88MW7). The **tether-formation step is COVERED**: PP_2320 is a high-confidence ortholog (**78.9% identity over a 303-residue alignment**) of *P. aeruginosa* PA2854, experimentally shown to covalently anchor the OM

lipoprotein **OprI** to PG in live cells (PMID: 42100858). KT2440 encodes a co-syntenic OprI ortholog, **PP_2322** (Q88KG8, 83 aa, OprI/Braun's-lipoprotein family, Pfam PF11839), only ~2 genes from PP_2320 — an enzyme+substrate cassette conserved from PAO1.

The **tether-hydrolysis (release) step is a GAP / candidate_uncertain**. The reference release enzyme is *E. coli* DpaA/LdtF (P0AA99), a PF03734 murein L,D-endopeptidase that detaches Braun's lipoprotein from PG (PMID: 33947763). No KT2440 protein — and, on present evidence, no *Pseudomonas* protein — is a confident DpaA ortholog. The only fold-level fallback is PP_1451, a minimal (184 aa) catalytic-domain-only L,D-transpeptidase whose closest match is *P. aeruginosa* PA3756, not DpaA. Whether release in *Pseudomonas* is carried by a divergent enzyme, by PP_1451 moonlighting, or is simply not part of this organism's biology is the central open question. All conclusions are homology/syteny-based (strong genus transfer for formation; unresolved for release); **no direct KT2440 experiment exists**.

Key Findings

Finding 1 — The 23 KEGG ppu00550 candidates are entirely out of module scope (F001)

A proteome-wide scan of UP000000556 for the defining YkuD/ErfK L,D-transpeptidase catalytic domain (Pfam **PF03734** / InterPro **IPR005490**) returns **exactly two proteins — PP_2320 and PP_1451 — neither of which is in the candidate list**. Every one of the 23 supplied candidates is instead a canonical PG-biosynthesis/division protein: cytoplasmic Mur ligases and precursor enzymes (*murA* PP_0964, *murB* PP_1904, *murC* PP_1338, *murD* PP_1335, *murE* PP_1332, *murF* PP_1333, *mraY* PP_1334, *murG* PP_1337, *ddlA* PP_4346, *ddlB* PP_1339); carrier-lipid metabolism (*uppS* PP_1595, *uppP* PP_2862); penicillin-binding proteins performing 4→3 D,D-transpeptidation and glycan polymerization (*ftsI*/PBP3 PP_1331, *mrdA-I* PP_3741, *mrdA-II* PP_4807, *mrcA*/PBP1a PP_5084, *mrcB*/PBP1b PP_4683, *pbpC* PP_0572, *mtgA* PP_5107); SEDS glycosyltransferases (*ftsW* PP_1336, *mrdB*/RodA PP_4806); and D,D-carboxypeptidases (*dacA* PP_4803, *dacB* PP_2098). These share the substrate "peptidoglycan" with the module but no enzymes, no EC classes, and — crucially — not the cysteine-active-site YkuD fold that defines the module. This is the definitive evidence that the candidate list, though a correct enumeration of KEGG "Peptidoglycan biosynthesis," does not represent this module.

Finding 2 — PP_2320 is the ErfK-family lipoprotein-anchoring L,D-transpeptidase; substrate = OprI/PP_2322 (F002, F005, F006, F007)

PP_2320 (Q88KH0, 325 aa) is the clear ortholog of experimentally validated *P. aeruginosa* PAO1 PA2854 (Q9HZZ0, 323 aa). Quantitatively, Smith–Waterman over full sequences gives **78.9% identity across a 303-aa alignment** — orthology-grade similarity (calibration: *E. coli* LdtA vs LdtB = 67% over 285 aa). PA2854 was directly shown to be "*the catalyst that performs this transformation between the cell wall and the outer-membrane lipoprotein OprI in live*" *P. aeruginosa* (PMID: 42100858). PP_2320 also aligns to the *E. coli* anchoring clade — 50.3% to LdtB/YbiS and 46.7% to LdtA/ErfK over the ~169-aa catalytic region — the very family defined as "*responsible for the attachment of the Braun lipoprotein to murein*" (PMID: 17369299). Its 4-mer Jaccard similarity to PA2854 (23.2%) exceeds that to any other *E. coli*/*Pseudomonas* L,D-transpeptidase ($\leq 3.3\%$), and it shows no homology to DpaA/LdtF (only an 8-aa spurious local alignment).

The substrate is present and co-syntenic. **OprI = PP_2322** (Q88KG8, 83 aa) belongs to the OprI/Braun's-lipoprotein family (Pfam **PF11839**, IPR021793), is a lipid-anchored lipoprotein with a signal peptide, and is a same-length ortholog of PAO1 OprI/PA2853 (P11221). In PAO1, *oprI* is immediately adjacent to the validated anchoring enzyme PA2854; in KT2440 the pairing is conserved — PP_2320 lies ~2 loci from *oprI*/PP_2322, separated only by the uncharacterized PP_2319 and a PP_2321 gap. Conserved enzyme–substrate synteny across two *Pseudomonas* species substantially strengthens transfer of PA2854's validated function to PP_2320.

Finding 3 — Domain architecture cements PP_2320 in the anchoring clade (F005)

PP_2320 carries a **LysM peptidoglycan-binding domain (IPR018392)** plus the **L,D-transpeptidase catalytic domain (IPR005490, residues 98–233)**, is classified in the "Bacterial L,D-transpeptidase" family **IPR050979**, and has an N-terminal signal peptide — an architecture identical to validated PA2854. Anchoring Ldts (*E. coli* LdtA/B/C) characteristically carry LysM PG-binding modules, which the release-clade DpaA lacks. This architectural signature independently corroborates the orthology-based assignment of PP_2320 to tether *formation*.

Finding 4 — The release enzyme (DpaA/LdtF) has no confident KT2440 ortholog (F003, F004, F008)

The *E. coli* Lpp-PG release enzyme is **DpaA/LdtF/YafK** = UniProt P0AA99 (246 aa, PF03734), a murein L,D-endopeptidase that "*catalyzes the cleavage of the cross-link between the outer membrane-anchored Braun's lipoprotein (Lpp) and peptidoglycan, detaching Lpp*" and whose title finding is "*DpaA Detaches Braun's Lipoprotein from Peptidoglycan*" (PMID: 33947763). Re-running 4-mer Jaccard against the correct DpaA sequence gives PP_1451 = 0.2% and PP_2320 = 0.0% — neither KT2440 YkuD protein groups with DpaA. Notably, **no** PAO1 YkuD protein groups with DpaA either (PA2854/PA3756/PA0732 all $\leq 0.7\%$), suggesting the release clade is absent or highly divergent across the genus. Because k-mer/Smith-Waterman metrics are insensitive to divergent homologs, this argues against a *close* ortholog but cannot by itself prove total absence — hence `candidate_uncertain` rather than a hard `not_expected`.

Finding 5 — PP_1451 is a second, functionally uncertain L,D-transpeptidase (F003, F006)

PP_1451 (Q88MW7, 184 aa) is the only other PF03734 protein in KT2440. It is **minimal** — catalytic domain only, **no LysM**, and not in IPR050979. Its best match is *P. aeruginosa* PA3756 (Q9HXN9, 166 aa) at 67.4% identity over 138 aa — an ortholog of undefined physiological role. It shows negligible similarity to *E. coli* DpaA/LdtF (only a 16-aa spurious alignment), to the anchoring clade, or to the 3→3 crosslinking clade (LdtD/E). It is therefore the only structural candidate for a release activity but is not, on sequence grounds, a DpaA ortholog. It should be flagged `candidate_uncertain` and resolved experimentally.

Finding 6 — Reduced L,D-transpeptidase complement (F002, F003)

KT2440 encodes only 2 PF03734 proteins versus 3 in *P. aeruginosa* PAO1 (PA2854, PA3756, PA0732) and 6 in *E. coli* K-12. The smaller complement is itself circumstantial support for the release-step gap: *P. putida* appears to have retained anchoring capacity while lacking a dedicated DpaA-clade release enzyme.

Target-Organism Pathway Definition

Included process (module scope). Covalent, reversible attachment of the C-terminus of a Braun-lipoprotein-like OM lipoprotein to the *meso*-diaminopimelate (mDAP) position of a PG tetrapeptide stem, catalyzed by an L,D-transpeptidase (acyl acceptor = protein rather than muropeptide), plus the reverse reaction — hydrolysis of that lipoprotein–mDAP amide bond by an LdtF/DpaA hydrolase, releasing the lipoprotein. In *E. coli* the lipoprotein is Braun's lipoprotein (Lpp); **in *Pseudomonas* the functional counterpart is OprI** (OprI/Braun's-lipoprotein family, Pfam PF11839). This is a **lineage-specific substrate substitution** curation must record explicitly.

Neighboring processes to keep separate.

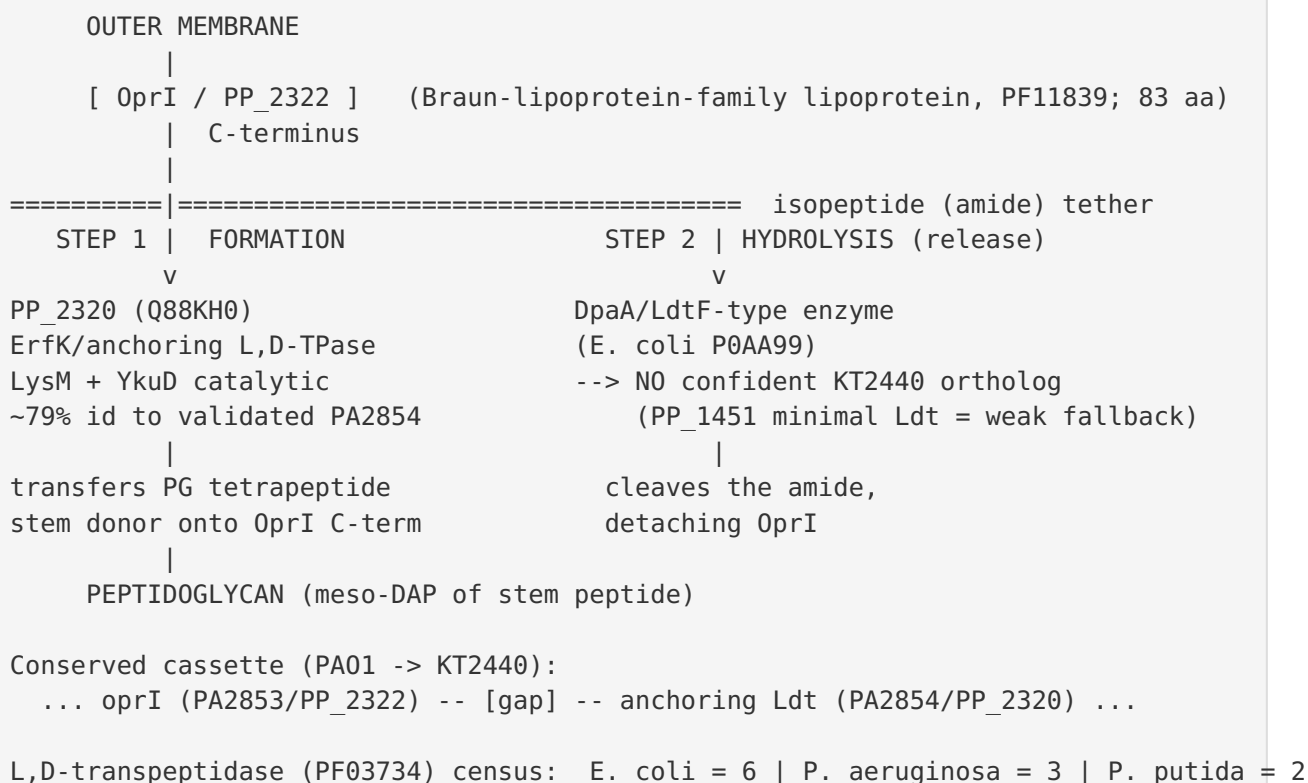
- KEGG [ppu00550](#) **Peptidoglycan biosynthesis** (Mur pathway, glycan polymerization, PBP 4→3 D,D-transpeptidation) — the source of the candidate list; *not* the module.
- Over-broad overview/amino-acid maps cross-listing individual genes (*murE*→ppu00300, *murD*→ppu00470, division maps ppu01501/01502).
- **3→3 L,D-transpeptidation** for PG crosslinking / carbapenem resistance (*E. coli* LdtD/LdtE) — same superfamily, **different reaction** (muropeptide acceptor).
- **Non-covalent** OM–PG tethering by **Pal/OprL** and by OmpA/β-barrel proteins ([PMID: 33139883](#)) — parallel mechanisms, not this covalent module.

Alternate names / database definitions. L,D-transpeptidase = "Ldt", "YkuD-domain protein", ErfK/YbiS/YcfS (*E. coli* LdtA/LdtB/LdtC); release enzyme = DpaA = LdtF = YafK. Relevant GO: **GO:0071972** (peptidoglycan L,D-transpeptidase activity); the tether-hydrolase activity is not cleanly represented in GO (see recommendations).

Expected Step Model and Satisfiability

Module step	Expected enzyme family	Diagnostic domain	KT2440 assignment	Verdict
1. Tether formation (lipoprotein → PG anchoring)	ErfK/anchoring-clade L,D-transpeptidase	PF03734 + LysM (IPR018392)	PP_2320 (Q88KH0) acting on OprI / PP_2322 (Q88KG8)	covered (high; strong genus transfer)
2. Tether hydrolysis (lipoprotein release)	YafK/LdtF/DpaA L,D-endopeptidase	PF03734 (DpaA clade)	No confident ortholog; weak fallback PP_1451 (Q88MW7)	candidate_uncertain / gap (moderate)

Mechanistic Model / Interpretation



Interpretation. *P. putida* KT2440 possesses a complete, syntenically organized machine for the **forward** (anchoring) reaction, using OprI as its Braun-lipoprotein surrogate. The **reverse** (release) reaction — catalyzed in *E. coli* by DpaA/LdtF — has no clear counterpart. Because the release enzyme belongs to the same PF03734 fold, the reduced YkuD complement of KT2440 (2 vs 6 in *E. coli*) is circumstantial evidence that a dedicated release function may be absent or repurposed in this lineage. Biologically this is plausible: OM tethering via OprI is a constitutive envelope feature, whereas regulated detachment (e.g., for division-site remodeling) may be dispensable or handled by a divergent enzyme in *Pseudomonas*.

Candidate Genes and Evidence (Detail)

The module's true genes (NOT in the KEGG candidate list)

Gene	UniProt	Length	Role	Key evidence	Curation status
PP_2320	Q88KH0	325 aa	ErfK-family OprI-anchoring L,D- transpeptidase	78.9% id/303 aa to validated PA2854 (P 42100858); 47–50% to E. coli anchoring Ldts (P 17369299); LysM+catalytic (IPR050979); syntenic to OprI	covered — promote
PP_2322 / OprI	Q88KG8	83 aa	Braun's- lipoprotein- family substrate (acyl acceptor)	PF11839/ IPR021793; ortholog of PAO1 OprI (P11221); OprI covalent attachment validated in PAO1 (P 42100858)	substrate — promote
PP_1451	Q88MW7	184 aa	Second, uncertain L,D- transpeptidase (release fallback)	Ortholog of PA3756 (67.4%/ 138 aa); no homology to DpaA/LdtF (≤ 16 aa); catalytic- only, no LysM	candidate_uncertain — promote

The 23 KEGG `ppu00550` candidates — assessment (all OUT OF SCOPE)

All 23 are canonical PG-metabolism genes and should be marked **not part of this module**; none carries PF03734.

- **Cytoplasmic Mur/Ddl/carrier steps** (murA–G, mraY, ddlA/B, uppS, uppP): standard, high-confidence PG-precursor biosynthesis; correct EC assignments; over-broad KEGG bucket tags (murE→ppu00300, murD→ppu00470) reflect map cross-listing, not different function.
- **PBP transpeptidases/glycosyltransferases** (ftsI/PBP3; mrdA-I, mrdA-II/PBP2; mrcA/PBP1a; mrcB/PBP1b; pbpC; mtgA; ftsW; mrdB/RodA): perform 4→3 D,D-transpeptidation and glycan polymerization — explicitly excluded by module scope. **mrdA-I/mrdA-II is a genuine PBP2 paralog pair** (annotation consistent, not over-propagation).
- **D,D-carboxypeptidases** (dacA, dacB): PG maturation, EC 3.4.16.4; not module members.
- **Broad EC/GO caveat**: EC 2.4.99.28 (ftsW, mrdB, mtgA, pbpC, mrcA, mrcB) and EC 3.4.16.4 (ftsI, mrdA-I/II, mrcA, dacA) are shared across many PBPs; these broad mappings must not be used to infer L,D-transpeptidase activity.

Gaps, Ambiguities, and Likely Over-Annotations

1. **Release step is the principal gap.** No KT2440 protein is a confident DpaA/LdtF ortholog; the whole *Pseudomonas* genus lacks a close DpaA ortholog by k-mer and Smith–Waterman metrics. Possibilities: (i) genuine lineage loss of a dedicated tether hydrolase; (ii) release by a divergent enzyme (PP_1451 or an uncharacterized peptidase); (iii) the OprI tether is not actively released in *P. putida*. Mark `candidate_uncertain`, **leaning gap**.
2. **PP_1451 ambiguity**: could be a minor anchoring Ldt, a 3→3 crosslinker, or a release enzyme — currently unassignable; needs full review.
3. **Candidate-list mismatch (`module_needs_revision` at the bucket level)**: binding this module to KEGG `ppu00550` mis-scopes it; the module's genes are not in that KEGG map. A **boundary error to flag**, not a biology gap.
4. **Generic UniProt names**: PP_2320 and PP_1451 are both "L,D-TPase catalytic domain-containing protein" — under-annotated relative to the evidence for PP_2320.
5. **Over-annotation risk**: none of the 23 KEGG genes should be propagated into the tether module on "peptidoglycan" keyword overlap.

Module and GO-Curation Recommendations

Module step	Recommended status	Evidence basis
Tether formation (anchoring)	covered	PP_2320 (79% to validated PA2854; LysM+catalytic; syntenic to OprI/PP_2322)
Tether hydrolysis (release)	candidate_uncertain → gap	No confident DpaA ortholog; PP_1451 weak fallback
KEGG <code>ppu00550</code> → module link	module_needs_revision (boundary error)	All 23 candidates canonical PG biosynthesis; no PF03734

Concrete actions:

1. **Detach** the module from KEGG bucket `ppu00550`; record the boundary rationale (no shared enzymes/chemistry). Add explicit exclusions: PBP D,D-transpeptidases, Mur ligases, Pal/OprL, OmpA/β-barrel tethers.
2. **Add** PP_2320 + PP_2322 as the tether-formation enzyme+substrate pair. Record OprI (PP_2322) as the **lineage-specific substrate** replacing enterobacterial Lpp.
3. **Register** the release step as `candidate_uncertain / gap`, noting PP_1451 as the sole fold-level fallback.
4. **GO annotation:** update PP_2320 to "ErfK-family lipoprotein (OprI)-anchoring L,D-transpeptidase" with **GO:0071972** (peptidoglycan L,D-transpeptidase activity) and an OprI-anchoring biological-process term; evidence code ISS/ISO from PA2854
(`P 42100858`)
+ IEA domain support.
5. **GO term request likely needed** for the lipoprotein–peptidoglycan tether hydrolase (DpaA) activity: current GO lacks a specific "outer-membrane-lipoprotein–peptidoglycan cross-link hydrolase" term; a dedicated term would support this module step across taxa.
6. **Do not** propagate the module to any of the 23 `ppu00550` genes.

Genes to Promote to Full fetch-gene Review

1. **PP_2320 (Q88KH0)** — promote; assign as OprI-anchoring L,D-transpeptidase (highest-value re-annotation).
2. **PP_1451 (Q88MW7)** — promote; resolve whether it is the (divergent) tether-release enzyme, a 3→3 crosslinker, or a minor anchoring Ldt.
3. **PP_2322 (Q88KG8, OprI)** — promote as the module substrate; confirm covalent-attachment status in *P. putida*.
4. (Lower priority) **PP_1223 (Pal/OprL)** — confirm it is only a *non-covalent* PG-associated lipoprotein, to keep it out of this module.

Evidence Base

PMID	Paper (abbrev.)	Role in this review
42100858	<i>Outer Membrane–Peptidoglycan Anchoring in Pseudomonas</i> (El-Araby et al. 2026)	Primary support. Directly shows PA2854 anchors OprI to PG in live <i>P. aeruginosa</i> ; validated function transferred to ortholog PP_2320. Verified: " <i>the gene product of PA2854 is the catalyst that performs this transformation between the cell wall and the outer-membrane lipoprotein OprI in live.</i> "
17369299	<i>Identification of the L,D-transpeptidases responsible for attachment of the Braun lipoprotein to E. coli peptidoglycan</i> (Magnet et al. 2007)	Defines the ErfK/Ldtfm family as the Braun-lipoprotein anchoring enzymes — the family of PP_2320. Verified: " <i>in Escherichia coli Ldt(fm) homologues are responsible for the attachment of the Braun lipoprotein to murein.</i> "
33947763	<i>DpaA Detaches Braun's Lipoprotein from Peptidoglycan</i> (Winkle et al. 2021)	Defines the release step (DpaA/LdtF, P0AA99) whose ortholog is missing/uncertain in KT2440.
33139883	<i>β-Barrel proteins tether the outer membrane in many Gram-negative bacteria</i> (Sandoz et al. 2021)	Context: alternative/parallel OM–PG tethering mechanisms distinct from this covalent module.

How the evidence combines. The formation step rests on *direct experimental* evidence in the sister species *P. aeruginosa* (PMID: 42100858) plus quantitative orthology (78.9% id/303 aa), anchoring-clade domain architecture, and conserved enzyme–substrate synteny — a strong, though not KT2440-direct, transfer. The release step rests only on *absence-of-ortholog* reasoning against the *E. coli* reference DpaA (PMID: 33947763), which is weaker (a divergent homolog could evade k-mer/Smith–Waterman detection).

Key database identifiers: PP_2320 Q88KH0, PP_1451 Q88MW7, PP_2322/OprI Q88KG8; PA2854 Q9HZZ0, OprI/PA2853 P11221; *E. coli* DpaA/LdtF P0AA99. Pfam PF03734, PF11839; InterPro IPR005490, IPR018392, IPR050979, IPR021793.

Limitations and Knowledge Gaps

1. **No direct KT2440 experiment.** Both the anchoring assignment and the release-step gap are computational (orthology, domain architecture, synteny). The strongest experimental anchor is in *P. aeruginosa*, not *P. putida*.
 2. **k-mer/Smith–Waterman insensitivity to divergent homologs.** The conclusion that KT2440 lacks a DpaA ortholog could be a false negative if the release enzyme is structurally conserved but sequence-divergent. Profile-HMM and structure-based searches were not exhausted.
 3. **PP_1451 function is genuinely unknown.** Its "fallback" status is by elimination, not positive evidence; it could be an anchoring enzyme, a release enzyme, or neither.
 4. **OprI-as-covalent-substrate in KT2440 is inferred**, not shown biochemically for this strain; it rests on orthology to PAO1 OprI.
 5. **Module GO vocabulary may be incomplete** — a specific lipoprotein–peptidoglycan cross-link hydrolase term may need to be requested.
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Proposed Follow-up Experiments / Actions

Computational (immediate, low cost): - Run **profile-HMM (HMMER/InterPro) and structure-based (Foldseek/DALI)** searches for a DpaA-clade protein across the KT2440 proteome and the *Pseudomonas* pangenome, to convert `candidate_uncertain` into a confident

gap or to recover a divergent release enzyme. - Build a phylogeny of all *Pseudomonas* PF03734 proteins with *E. coli* LdtA–F as references, to place PP_1451/PA3756 definitively. - AlphaFold-model PP_2320 and PP_1451; confirm the catalytic Cys and LysM placement in PP_2320.

Experimental (to resolve gaps): - **Detect the covalent OprI–PG species** in KT2440 by mucopeptide LC-MS/MS; test dependence on PP_2320 via a clean deletion (ΔPP_2320), directly confirming anchoring in the target strain. - **Test PP_1451** for L,D-endopeptidase / tether-release activity in vitro and via ΔPP_1451 mucopeptide profiling; assess whether OprI release accumulates or is lost. - **Complementation:** express *E. coli* *dpaA* in KT2440 to see whether an exogenous release enzyme alters OprI–PG turnover, testing the "release-not-required" hypothesis.

Curation (now): - Detach module from **ppu00550**; add PP_2320 + PP_2322; register release step as **candidate_uncertain** / **gap**; promote PP_2320, PP_1451, PP_2322 to full gene review; flag the OprI-for-Lpp lineage-specific substrate substitution.

Evidence-Type Summary

- **Direct experiment (target strain):** none identified for the module in *P. putida* KT2440.
- **Direct experiment (same genus, strong transfer):** PA2854→OprI anchoring in *P. aeruginosa* ([PMID: 42100858](#)); OprI orthology.
- **Homology + domain + synteny (target strain):** PP_2320 anchoring assignment; PP_2322 substrate.
- **Inferred negative / gap:** absence of a confident DpaA/LdtF release ortholog in KT2440.